

### Datasets in scope (D7 excluded — FAIL reads)

| Dataset      | Organism          | Chemistry | Reads  | Ref    | FAST5    |
|--------------|-------------------|-----------|--------|--------|----------|
| D1           | SARS-CoV-2        | R9.4      | 1,383  | 31 KB  | 11.00 GB |
| D2           | E. coli           | R9.4      | 89     | 5 MB   | 27.00 GB |
| D3           | S. cerevisiae     | R9.4      | 500    | 12 MB  | 0.40 GB  |
| D4           | C. reinhardtii    | R9.4      | 500    | 112 MB | 2.00 GB  |
| D6           | E. coli           | R10.4     | 294    | 5 MB   | 98.00 GB |
| RB_ecoli     | E. coli           | R10.4.1   | 12,000 | 5 MB   | 0.93 GB  |
| RB_dmel      | D. melanogaster   | R10.4.1   | 12,000 | 143 MB | 0.73 GB  |
| RB_hsap_full | H. sapiens (GRCh) | R10.4.1   | 12,000 | 3.0 GB | 7.10 GB  |

## Datasets — extended characterization

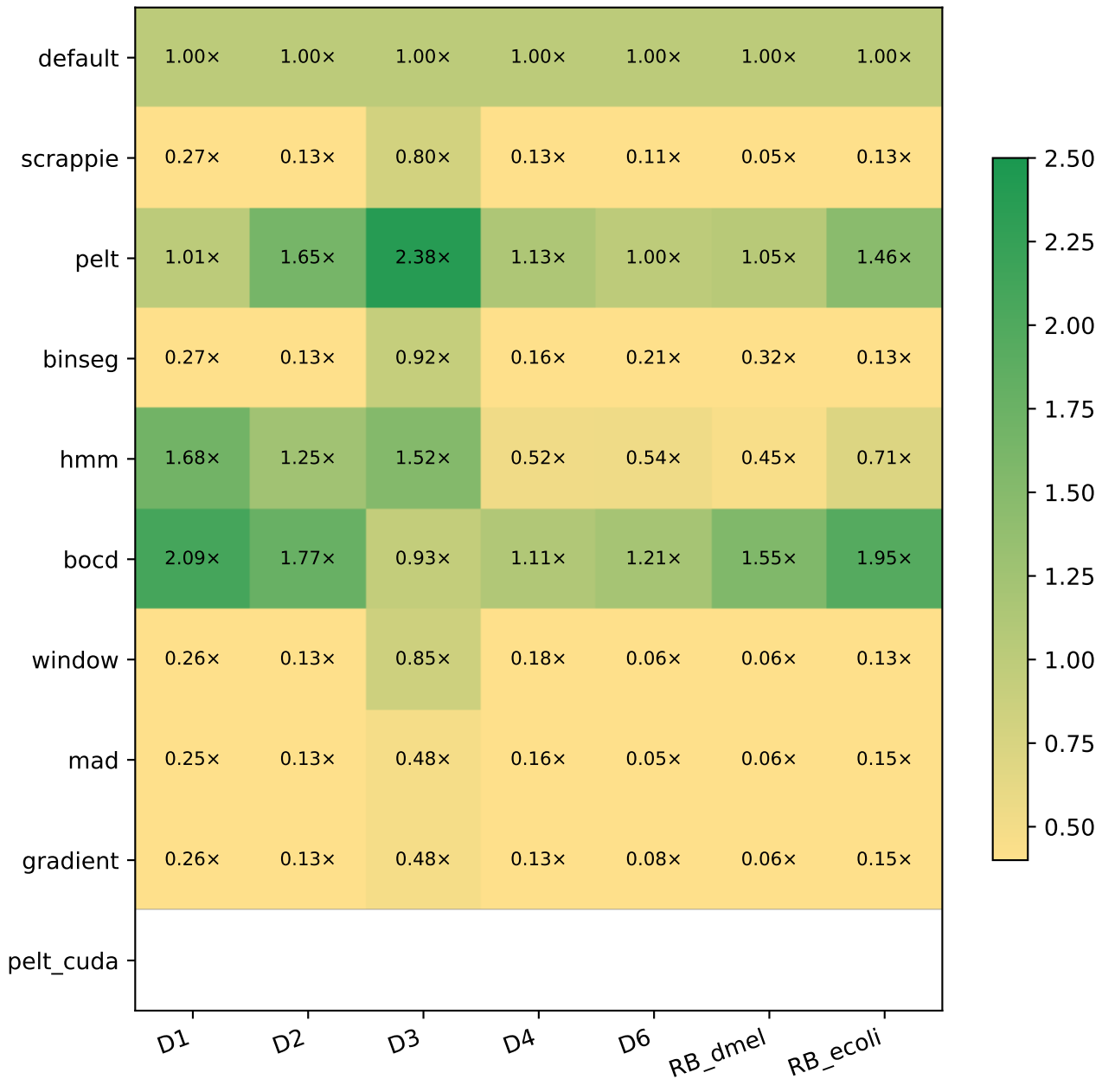
| DS       | Organism        | Chem    | Reads  | Ref bp    | GC%  | contigs | N50 (Mb) | N%   | masked% |
|----------|-----------------|---------|--------|-----------|------|---------|----------|------|---------|
| D1       | SARS-CoV-2      | R9.4    | 1,383  | 0.03 Mb   | 38.0 | 1       | 0.03     | 0.00 | 0.0     |
| D2       | E. coli         | R9.4    | 89     | 5.23 Mb   | 50.5 | 1       | 5.23     | 0.00 | 0.0     |
| D3       | S. cerevisiae   | R9.4    | 500    | 12.16 Mb  | 38.1 | 17      | 0.92     | 0.00 | 0.0     |
| D4       | C. reinhardtii  | R9.4    | 500    | 111.10 Mb | 64.1 | 53      | 7.78     | 3.65 | 32.4    |
| D6       | E. coli         | R10.4   | 294    | 5.23 Mb   | 50.5 | 1       | 5.23     | 0.00 | 0.0     |
| RB_dmel  | D. melanogaster | R10.4.1 | 12,000 | 143.73 Mb | 42.0 | 1,870   | 25.29    | 0.80 | 20.9    |
| RB_ecoli | E. coli         | R10.4.1 | 12,000 | 5.23 Mb   | 50.5 | 1       | 5.23     | 0.00 | 0.0     |

*Ref bp* = total reference length; *N50* = contig N50; *N%* = ambiguous N bases; *masked%* = soft-masked low-complexity / repeats (lowercase in FASTA).

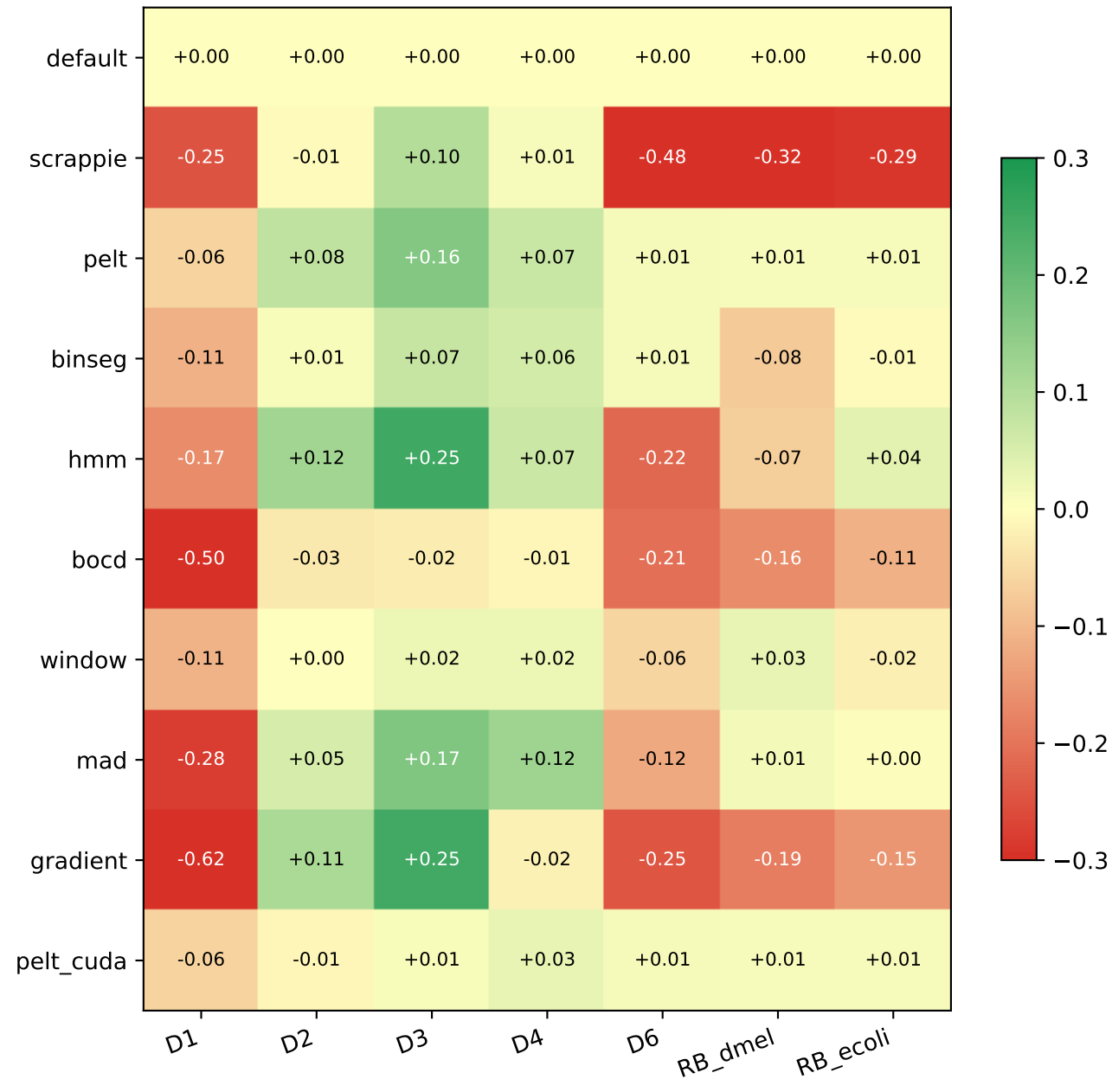
**F1 across (segmenter, dataset) — best CV-tuned config used for HMM/BOCD**



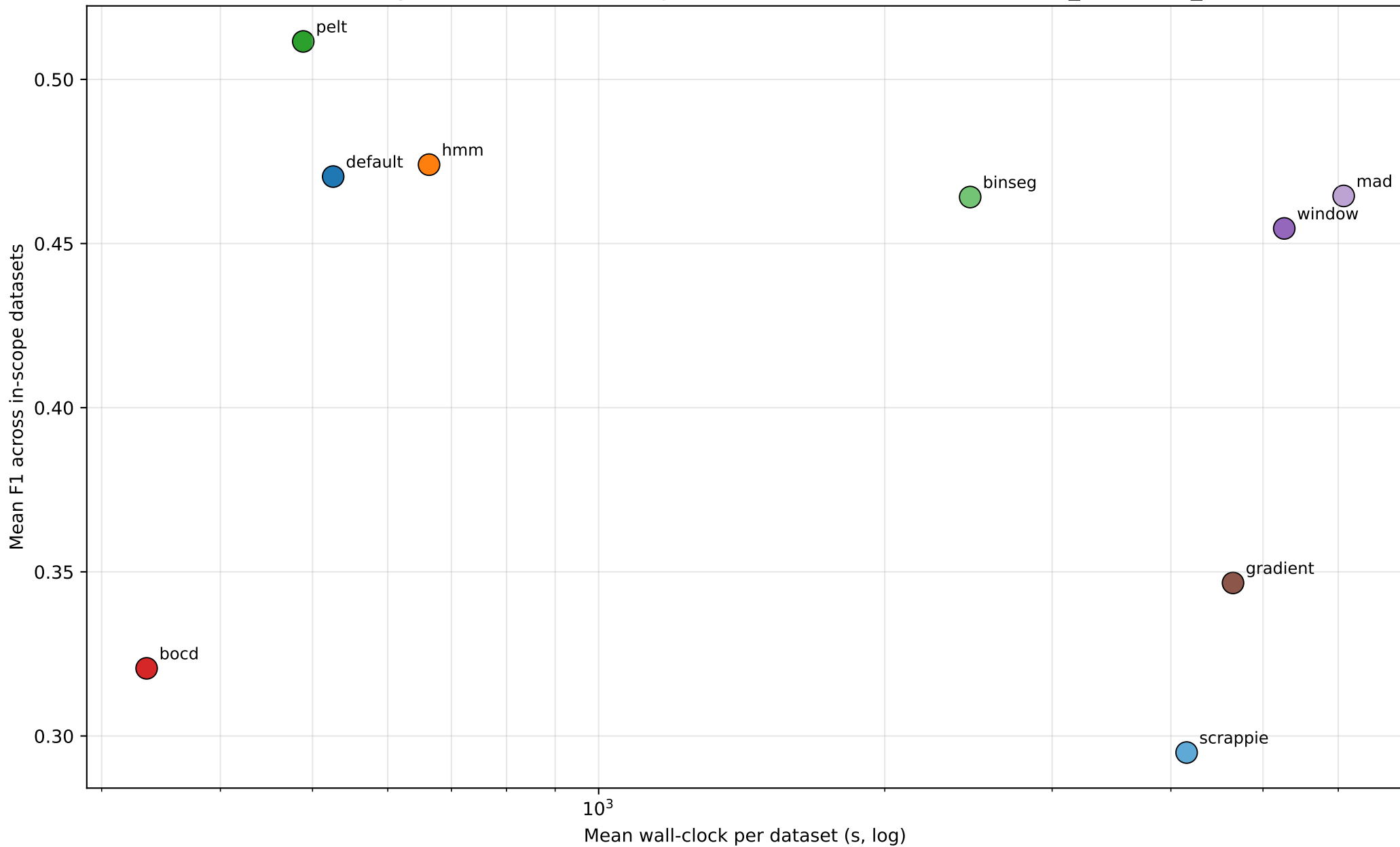
Speedup vs default



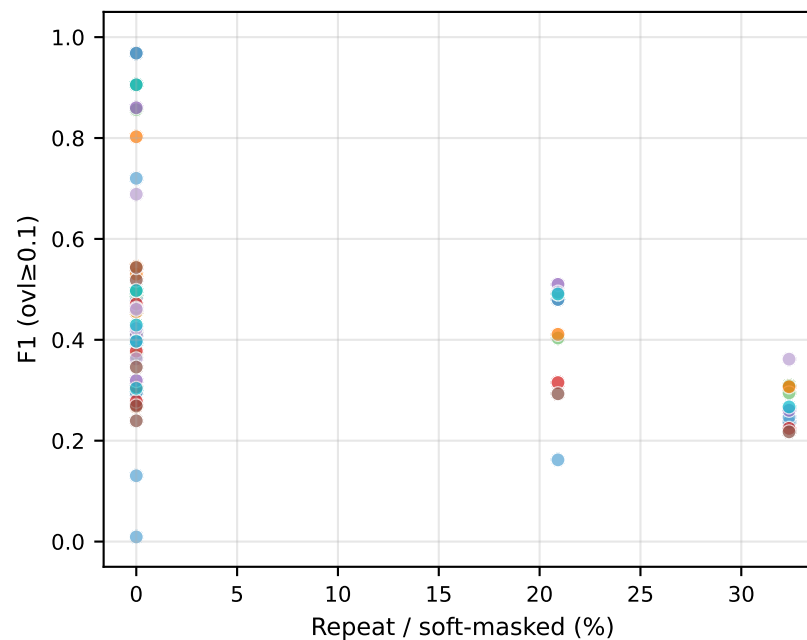
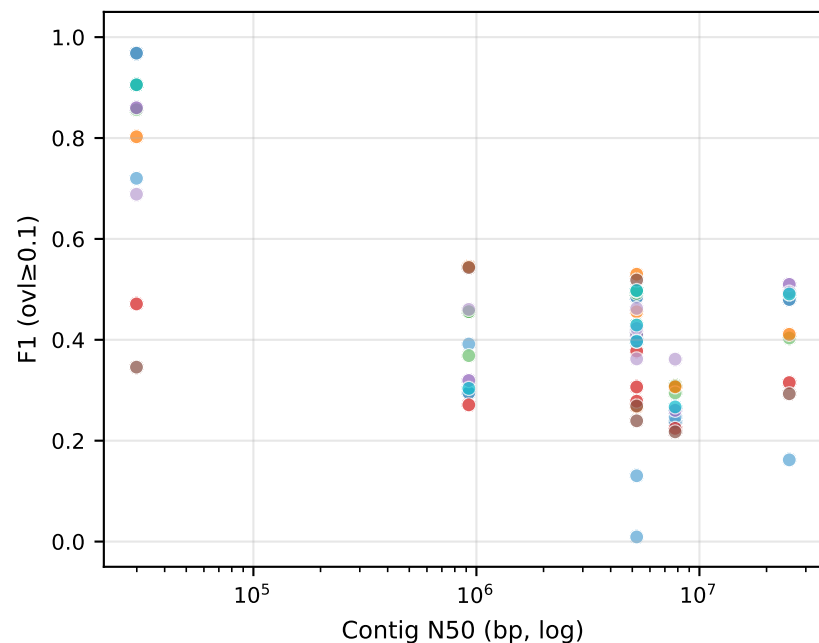
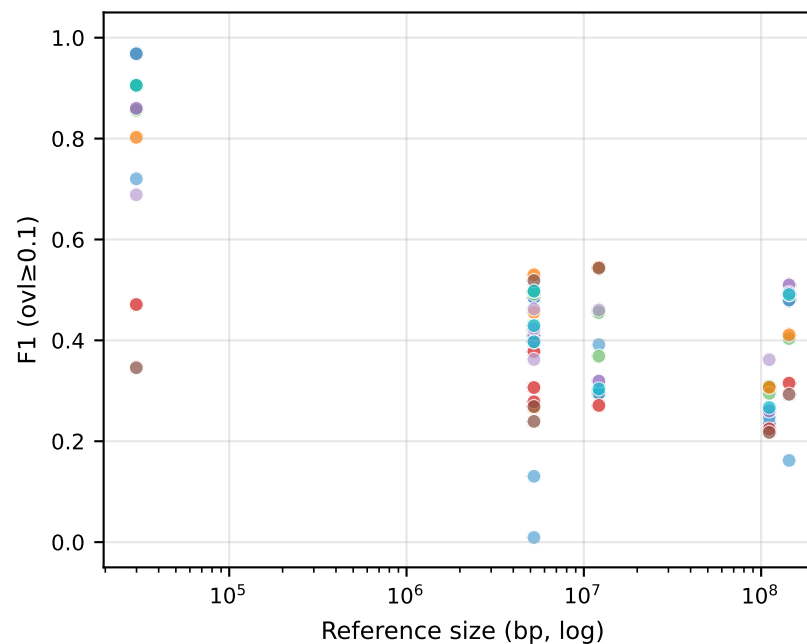
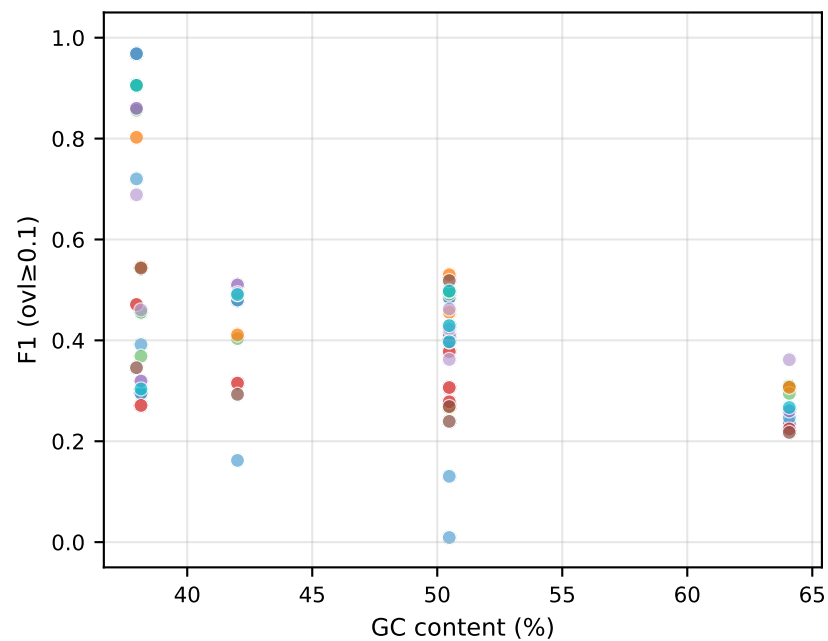
F1 delta vs default



Pareto: quality vs runtime (averaged across D1,D2,D3,D4,D6,RB\_dmel,RB\_ecoli)



# F1 vs reference attributes — per segmenter (Setup A, $ovl \geq 0.1$ )

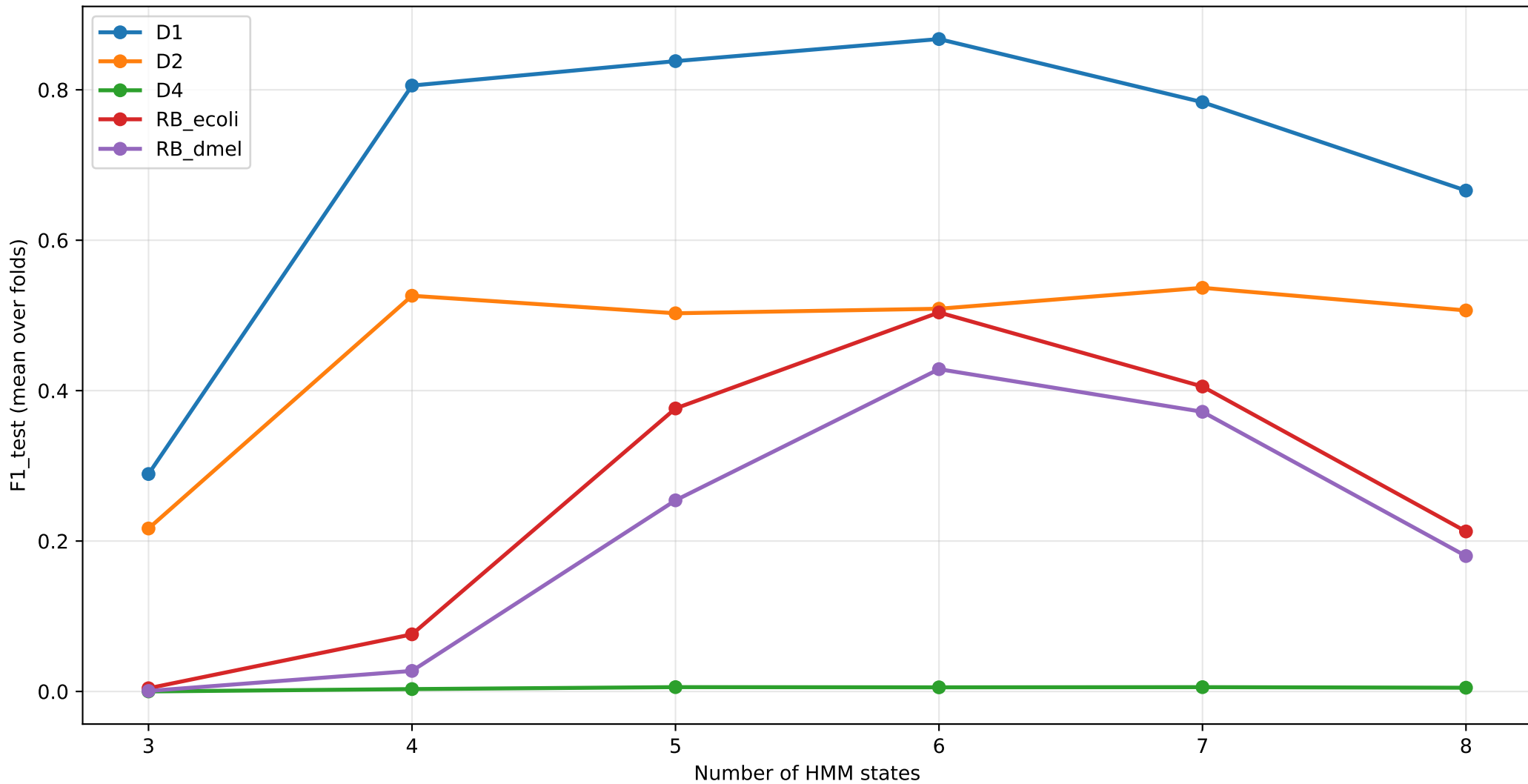


- default
- scrappie
- pelt
- binseg
- hmm
- bocd
- window
- mad
- gradient
- pelt\_cuda

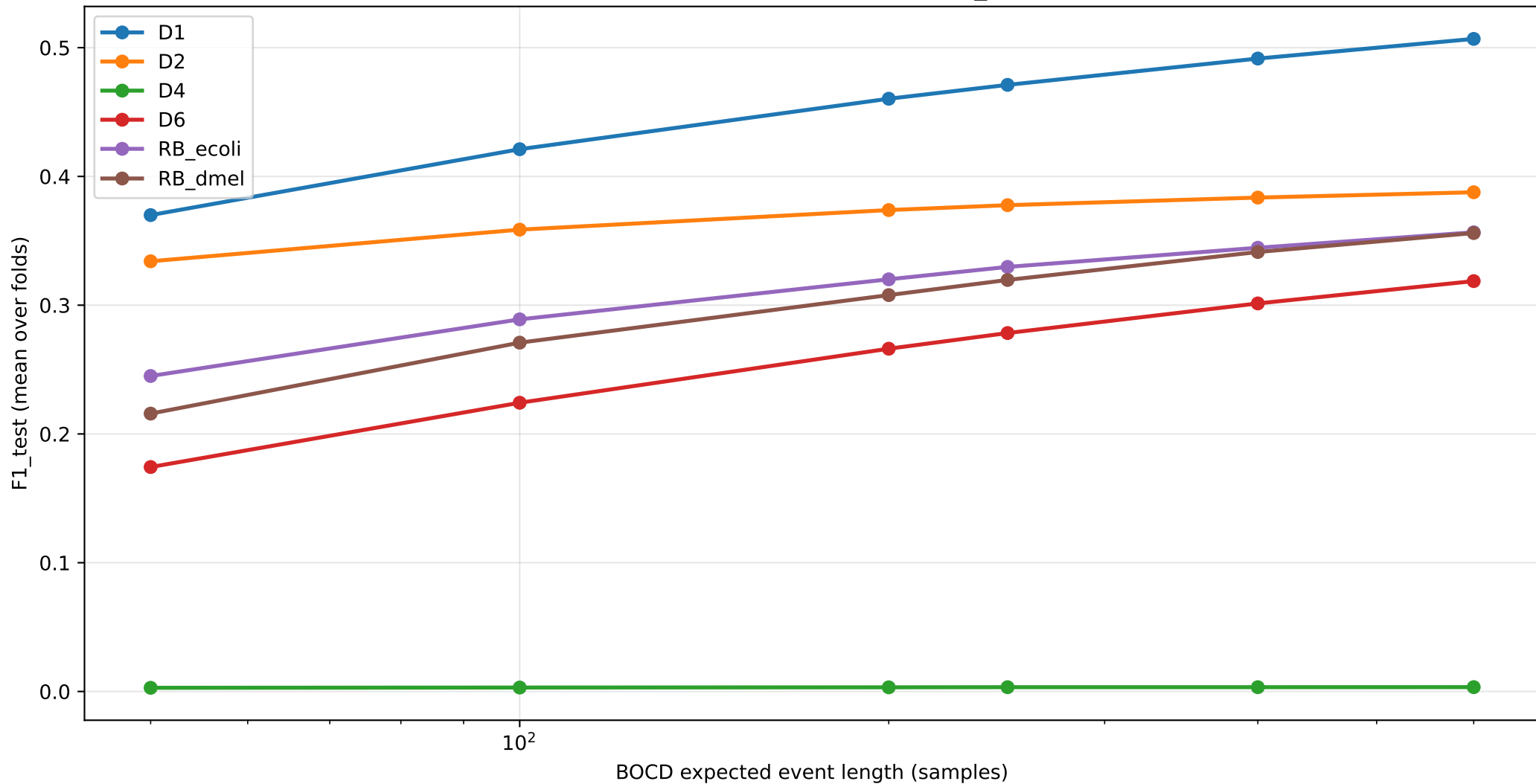
## Pearson correlation: F1 (ovl $\geq$ 0.1) vs reference attribute



**HMM CV sweep: F1 vs STATES (STAY=0.96)**



BOCD CV sweep: F1 vs EXP\_LEN



## Per-dataset segmenter recommendation (BEST F1 vs FAST-near-default)

| Dataset  | default (F1   wall)     | BEST F1 segmenter              | FAST + $\geq 95\%$ F1 segmenter |
|----------|-------------------------|--------------------------------|---------------------------------|
| D1       | default (0.968   1359s) | default (0.968   1359s) [1.00× | default (0.968   1359s) [1.00×  |
| D2       | default (0.410   549s)  | hmm (0.530   439s) [1.25×      | hmm (0.530   439s) [1.25×       |
| D3       | default (0.295   6s)    | hmm (0.545   4s) [1.52×        | hmm (0.545   4s) [1.52×         |
| D4       | default (0.237   13s)   | mad (0.362   83s) [0.16×       | mad (0.362   83s) [0.16×        |
| D6       | default (0.484   1332s) | pelt (0.498   1337s) [1.00×    | default (0.484   1332s) [1.00×  |
| RB_dmel  | default (0.480   366s)  | window (0.510   6401s) [0.06×  | pelt (0.491   350s) [1.05×      |
| RB_ecoli | default (0.419   55s)   | hmm (0.456   76s) [0.71×       | hmm (0.456   76s) [0.71×        |

## default vs tuned-pelt — wall-time comparison

| Dataset  | default wall | pelt wall | speedup | F1 default | F1 pelt |
|----------|--------------|-----------|---------|------------|---------|
| D1       | 1359.1s      | 1350.9s   | 1.01×   | 0.9682     | 0.9056  |
| D2       | 549.1s       | 333.6s    | 1.65×   | 0.4097     | 0.4923  |
| D3       | 5.8s         | 2.5s      | 2.38×   | 0.2951     | 0.4558  |
| D4       | 13.0s        | 11.5s     | 1.13×   | 0.2366     | 0.3091  |
| D6       | 1331.8s      | 1336.5s   | 1.00×   | 0.4845     | 0.4976  |
| RB_dmel  | 366.1s       | 350.0s    | 1.05×   | 0.4799     | 0.4913  |
| RB_ecoli | 54.6s        | 37.3s     | 1.46×   | 0.4188     | 0.4292  |